SPECTRAL INVARIANTS

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Abstract. This is a note for the lectures I gave in the course Symplectic Geometry WS2024/2025 at the Universität Augsburg.

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1. Lusternik–Schnirelmann spectral invariants

Let M be a closed, connected, oriented smooth manifold of finite dimension. Given a $f \in C^1(M)$ and for any $a \in \mathbb{R} \setminus \mathrm{Spec}(f)$, denote by $M^{<a} = \{x \in M : f(x) < a\}$ the sublevel set. The inclusion $M^a \subset M$ leads to a long exact sequence:

$$\cdots \to H_*(M^{<a}) \xrightarrow{i^a} H_*(M) \xrightarrow{j^a} H_*(M^{\geq a}) \to \cdots$$

where

$$H_*(M^{\geq a}) \cong H_*(M, M^{<a}).$$

Let $A \in H_*(M)$ be a non-zero singular homology class, the *Lusternik–Schnirelmann spectral invariant* or *critical values selector* is defined as

$$\begin{aligned} c(A, f) &:= \inf_{\substack{\sigma = \sum_i n_i \sigma_i \in C_*(M) \\ [\sigma] = A}} \max_{\substack{\mathrm{supp}(\sigma) \\ = \cup_i \mathrm{im}(\sigma_i) \\ n_i \neq 0}} f \\ &= \inf\{a \in \mathbb{R} \setminus \mathrm{Spec}(f) : A \in \mathrm{im}(i^a)\} \\ &= \inf\{a \in \mathbb{R} \setminus \mathrm{Spec}(f) : A \in \ker(j^a)\} \end{aligned}$$

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Example 1.1. Let $M = T^2$ and f be a height function so that $\mathrm{Crit}(f) = \{p_{\min}, q_1, q_2, p_{\max}\}$ with $\min f = f(p_{\min}) < f(q_1) < f(q_2) < f(p_{\max}) = \max f$. Pick $[p_{\min}], [S^1 \times \{0\}], [\{0\} \times S^1], [T^2] \in H_*(T^2)$, then we have:

$$\begin{aligned} c([p_{\min}], f) &= f(p_{\min}) \\ c([S^1 \times \{0\}], f) &= f(q_1) \\ c([\{0\} \times S^1], f) &= f(q_2) \\ c([T^2], f) &= f(p_{\max}) \end{aligned}$$

Proposition 1.2 ([BHS21, Proposition 5.2])**.** *The LS spectral invariant c satisfies the following properties.*

(1) *c is invariant under diffeomorphism φ of M, i.e. $c(\varphi_* A, \varphi_* f) = c(A, f)$.*
(2) *$c(A, f) \in \mathrm{Spec}(f)$. (spectrality)*
(3) *$c([\mathrm{pt}], f) = \min f$, $c([M], f) = \max f$.*
(4) *$\forall A \in H_*(M) \setminus \{0\}$, $\forall f, g \in C^1(M)$:*

$$\min(f - g) \leq c(A, f) - c(A, g) \leq \max(f - g)$$

In particular, $c(A, \cdot) : C^1(M) \to \mathbb{R}$ is Lipschitz continuous w.r.t. C^0-norm, i.e.

$$|c(A, f) - c(A, g)| \leq \|f - g\|_{C^0}.$$

(5) *$c(A \cap B, f) \leq c(A, f)$ for any $B \in H_*(M)$ s.t. $A \cap B \neq 0$.*
(6) *Suppose $\deg(B) < \deg([M]) = \dim M$ and $c(A \cap B, f) = c(A, f)$. Then $f^{-1}(c(A, f))$ is homologically non-trivial, i.e. for any open neighborhood U of $f^{-1}(c(A, f))$, there is some $i > 0$ such that $H_i(U) \to H_i(M)$ is non-trivial. In particular, the set $f^{-1}(c(A, f))$ is (uncountably) infinite.*

Proof. **(1)** This is due to the following commutative diagram:

$$\begin{array}{ccc} H_*(\{f \geq a\}) & \xrightarrow{\varphi_*} & H_*(\{\varphi_* f \geq a\}) \\ {\scriptstyle j_f^a}\uparrow & & \uparrow{\scriptstyle j_{\varphi_* f}^a} \\ H_*(M) & \xrightarrow[\varphi_*]{} & H_*(M) \end{array}$$

(2) Suppose a is a regular value, then for sufficiently small $\varepsilon > 0$, $\mathrm{Crit}(f) \cap [a - \varepsilon, a + \varepsilon] = \varnothing$, and thus

$$H_*(M^{<a+\varepsilon}, M^{<a-\varepsilon}) \cong \{0\}.$$

From the long exact sequence

$$\cdots \to H_*(M^{<a-\varepsilon}) \to H_*(M^{<a+\varepsilon}) \to H_*(M^{<a+\varepsilon}, M^{<a-\varepsilon}) \to \cdots$$

we conclude that

$$H_*(M^{<a-\varepsilon}) \cong H_*(M^{<a+\varepsilon}).$$

We assume $a = c(A, f) \notin \mathrm{Spec}(f)$, then $A \in \mathrm{im}(i^{a+\varepsilon})$ and $A \notin \mathrm{im}(i^{a-\varepsilon})$. This contradicts to the above isomorphism. Hence, the assumption is false.
(3) These can be seen directly from the definition of c.
(4) Let $a \in \mathbb{R} \setminus \mathrm{Spec}(f)$. If $f < a$, then

$$\min(f - g) < f - g < a - g$$

which yields the inclusions $\{f < a\} \subset \{g < a - \min(f-g)\} \subset M$ and induce the commutative diagram:

$$H_*(\{f<a\}) \longrightarrow H_*(\{g < a-\min(f-g)\}) \xrightarrow{i_g^{a-\min(f-g)}} H_*(M)$$
$$\qquad\qquad\qquad

and the commutative diagram:

$$\begin{array}{ccc} H^{\dim M-\deg B}(M) & \xrightarrow{\sim} & H_{\deg B}(M) \\ \downarrow & & \downarrow \\ H^{\dim M-\deg B}(K) & \xrightarrow[\sim]{} & H_{\deg B}(M, M\setminus K) \end{array}$$

Since the right vertical arrow is non-trivial, we can deduce from above that $H^{\dim M-\deg B}(M) \to H^{\dim M-\deg B}(K)$ is non-trivial, hence its algebraic dual $H_{\dim M-\deg B}(K) \to H_{\dim M-\deg B}(M)$ is also non-trivial. This further factors through $H_{\dim M-\deg B}(U)$. It follows that $H_i(U) \to H_i(M)$ is non-trivial for some $i = \dim M - \deg B > 0$ since $\deg(B) < \dim M$.

□

Recall the *cup-length* is defined by

$$cl(M) := \max\{k : \exists \alpha_1, \dots, \alpha_k \in H^*(M), \deg(\alpha_i) > 0 \forall i, \alpha_1 \smile \cdots \smile \alpha_k \neq 0\}.$$

In terms of intersection product on homology by Poincaré duality, $A_i = \mathrm{PD}(\alpha_i)$, we equivalently have

$$cl(M) = \max\{k : \exists A_1, \dots, A_k \in H_*(M), \deg(A_i) < \dim M \forall i, A_1 \cap \cdots \cap A_k \neq 0\}.$$

The following statement is originally due to Lusternik-Schnirelmann theorem plus cup-length estimate.

Theorem 1.3. *For every $f \in C^1(M)$ on a closed connected oriented smooth manifold M:*

$$\#\operatorname{Crit}(f) \geq cl(M) + 1$$

Proof. Let $k = cl(M)$ and $A_1, \dots, A_k \in H_*(M)$ with $\deg(A_i) < \dim \forall i$ and $A_1 \cap \cdots \cap A_k \neq 0$. By Proposition 1.2 (5), we have:

$$c(A_1 \cap \cdots \cap A_k, f) \leq \cdots \leq c(A_1 \cap A_2, f) \leq c(A_1, f) \leq c([M], f) = \max f$$

Assume there is a $f \in C^1(M)$ s.t. $\#\operatorname{Crit}(f) \leq cl(M) = k$. Then at least one of the above inequalities must be an equality. This implies that there exist $A, B \in H_*(M)$ with $\deg(B) < \dim M$ such that $c(A \cap B, f) = c(A, f) =: c$. The set $f^{-1}(c)$ is under the assumption a finite set with cardinality $\leq k$. However, Proposition 1.2 (6) tells that in such case $f^{-1}(c)$ is homologically non-trivial and thus infinite. We get a contradiction. Hence, the assumption is false and the statement follows. □

There is a Morse theoretical approach to LS spectral invariants where all above results follow similarly for Morse functions. The spectral invariant for Morse function f is defined by

$$c(A, f) := \inf_{\substack{\sum_x n_x x \in CM_*(f) \\ [\sum_x n_x x] = A}} \max_{\substack{x \in \operatorname{Crit}(f) \\ n_x \neq 0}} f(x)$$

where we have applied the natural isomorphism

$$HM_*(f) \cong H_*(M)$$

to identify any homology class $A \in H_*(M;\mathbb{Z}_2)$ to a class $[\sum_{x\in \mathrm{Crit}(f)} a_x x] \in HM_*(f)$. The filtered Morse complexes read

$$CM_*^{<a}(f) := \bigoplus_{\substack{x\in \mathrm{Crit}(f)\\ f(x)<a}} \mathbb{Z}_2\langle x\rangle \subset CM_*(f)$$
$$CM_*^{>a}(f) := CM_*(f)/CM_*^{<a}(f) \cong \bigoplus_{\substack{x\in \mathrm{Crit}(f)\\ f(x)>a}} \mathbb{Z}_2\langle x\rangle$$

It yields long exact sequence on homology

$$\cdots HM_*^{<a}(f) \xrightarrow{i^a} HM_*(f) \xrightarrow{j^a} HM_*^{>a}(f) \to \cdots$$

which is isomorphic to the long exact sequence of singular homology above by the isomorphisms

$$HM_*^{<a}(f) \cong H_*(M^{<a})$$
$$HM_*^{>a}(f) \cong H_*(M, M^{>a})$$

The spectral invariant can be equivalently expressed as:

$$\begin{aligned} c(A,f) &= \inf\{a\in\mathbb{R}\setminus \mathrm{Spec}(f) : A \in \mathrm{im}(i^a)\} \\ &= \inf\{a\in\mathbb{R}\setminus \mathrm{Spec}(f) : A \in \ker(j^a)\} \end{aligned}$$

In particular, due to the C^0-Lipschitz continuity of the spectral invariant and the genericity of Morse functions in $C^2(M)$, one can extend the spectral invariant from Morse functions to any C^2-smooth functions on M recovering the notion we have before (in $\mathbb{Z}_2$ coefficient). C.f. [FN20, §11].

The generalization to infinite dimensional setup is in general difficult. It is often required that f satisfies certain boundedness property, e.g. Palais–Smale condition. C.f. [Cha93, Chapter II, §3.2].

2. Hamiltonian spectral invariants

Let (M^{2n},ω) be a compact, connected, symplectically aspherical manifold. When $\partial M \neq \varnothing$, we assume ∂M is a contact manifold and (M,ω) is a Liouville domain. We first discuss the $\partial M = \varnothing$ case in detail and give complementary descriptions for $\partial M \neq \varnothing$ case afterwards. Suppose $H \in C^\infty(S^1\times M)$ with $S^1 = \mathbb{R}/\mathbb{Z}$, the Hamiltonian action functional is defined on $C^\infty_{\mathrm{contr.}}(S^1, M)$ by

$$\mathcal{A}_H(x) = \int_{D^2} \bar{x}^*\omega - \int_{S^1} H_t(x(t))dt$$

where $\bar{x} : D^2 \to M$ is some cap of $x : S^1 \to M$. With the convention $dH = \omega(\cdot, X_H)$ and $g_J = \omega(\cdot, J\cdot)$, the positive L^2-gradient flow equation of $\mathcal{A}_H$ gives rise to the Floer equation

$$\partial_s u + J_t(u)(\partial_t - X_H(u)) = 0$$

for $u : \mathbb{R}\times S^1 \to M$. The finite energy solutions have asymptotics to 1-periodic Hamiltonian orbits x's which are critical points of $\mathcal{A}_H$, i.e. $\lim_{s\to\pm\infty} = x_\pm$ with $\dot{x}_\pm(t) = X_H(x_\pm(t))$ and $x_\pm(t+1) = x_\pm(t)$. By choosing generic pair $\{(H_t, J_t)\}_{t\in S^1} \subset$

$\mathcal{H}_{\mathrm{reg}} \times \mathcal{J}_{\mathrm{reg}}$, which in particular means H_t is non-degenerate, the Floer complex is well-defined with chain vector space

$$CF_*(H) = \bigoplus_{x \in \mathrm{Crit}(\mathcal{A}_H)} \mathbb{Z}_2\langle x \rangle,$$

graded by Conley–Zehnder index, i.e. $\deg x = \mu_{\mathrm{CZ}}(x)$ which takes value in $\mathbb{Z}$ when $c_1(M)|_{\pi_2(M)} = 0$ (otherwise only valued in $\mathbb{Z}/2\mathbb{Z}$ in general), and differential given by counting rigid Floer trajectories

$$\partial x = \sum_y \#_{\mathbb{Z}_2} \mathcal{M}^0(y, x) y$$

where $\mathcal{M}(y, x)$ is the moduli space of unparameterized Floer trajectories with asymptotics x and y, whose dimension is

$$\dim \mathcal{M}(y, x) = \deg x - \deg y - 1.$$

The Floer homology is

$$HF_*(H) = \frac{\ker \partial_*}{\operatorname{im} \partial_{*+1}}.$$

Denote the set of critical values of $\mathcal{A}_H$ by $\mathrm{Spec}(H)$ which is discrete subset of $\mathbb{R}$ (c.f. [Sch00, Proposition 3.7]), for any $a \in \mathbb{R} \setminus \mathrm{Spec}(H)$, we have filtered Floer subcomplex

$$CF_*^{<a}(H) = \bigoplus_{\substack{x \in \mathrm{Crit}(\mathcal{A}_H) \\ \mathcal{A}_H(x) < a}} \mathbb{Z}_2\langle x \rangle$$

and this leads to the filtration long exact sequence

$$\cdots \to HF_*^{<a}(H) \xrightarrow{i^a} HF_*(H) \xrightarrow{j^a} HF_*^{>a}(H) \to \cdots$$

Making use of the natural isomorphism

$$HM_{*+n}(f) \xrightarrow[\cong]{\Phi_{f,H}} HF_*(H) \xrightarrow[\cong]{\Phi_{H,f}} H_{*+n}(f),$$

the *Hamiltonian spectral invariant* ot *action selector* is defined as follows: for any $A \in H_*(M)$,

$$\begin{aligned} c(A, H) :=& \inf_{\substack{\sum_x n_x x \in CF_*(H) \\ [\sum_x n_x x] = A}} \max_{\substack{x \in \mathrm{Crit}(\mathcal{A}_H) \\ n_x \neq 0}} \mathcal{A}_H(x) \\ =& \inf\{a \in \mathbb{R} \setminus \mathrm{Spec}(H) : A \in \operatorname{im}(\Phi_{H,f} \circ i^a)\} \\ =& \inf\{a \in \mathbb{R} \setminus \mathrm{Spec}(H) : A \in \ker(j^a \circ \Phi_{f,H})\} \end{aligned}$$

Theorem 2.1 ([Sch00])**.** *The spectral invariant c satisfies the following properties.*

(1) *(spectrality) $\forall A \in H_*(M)$, $\forall H \in C^\infty(S^1 \times M)$: $c(A, H) \in \mathrm{Spec}(H)$.*

(2) *$\forall A \in H_*(M)$, $\forall H, K \in C^\infty(S^1 \times M)$:*

$$\int_0^1 \inf_{x \in M} (H_t(x) - K_t(x)) dt \leq c(A, H) - c(A, K) \leq \int_0^1 \sup_{x \in M} (H_t(x) - K_t(x)) dt$$

In particular, $c(A, \cdot) : C^\infty(S^1 \times M) \to \mathbb{R}$ is Lipschitz continuous w.r.t. Hofer norm and C^0-norm. The Hofer norm is defined by

$$\|H\| := \int_0^1 \left(\sup_{x \in M} H_t(x) - \inf_{x \in M} H_t(x) \right) dt.$$

(3) $\forall \varphi \in \mathrm{Symp}(M,\omega)$: $c(\varphi_* A, \varphi_* H) = c(A,H)$
(4) *If H is C^2-small, then $c(A,H)$ coincides with the LS spectral invariant.*
(5) *(triangle inequality)* $\forall H, K \in C^\infty(S^1 \times M)$, $\forall A, B \in H_*(M)$:

$$c(A \cap B, H\#K) \leq c(A,H) + c(B,K)$$

where $H_t \# K_t = H_t + K_t \circ (\varphi_H^t)^{-1}$.

Remark 2.2. The Hamiltonian spectral invariant is first defined for non-degenerate (i.e. regular) Hamiltonians, and due to the C^0-Lipschitz continuity, it gets extended to $C^\infty(S^1 \times M)$ fully.

The proof of these properties are more or less similar to those in terms of the Morse homology, which we will omit here. Despite the triangle inequality is somewhat more special: it involves the ring structure of Floer homology. To discuss this, it is more convenient to write the Floer equation in the following way: for $u : (\Sigma, j) \to (M, J)$ where Σ is a puncture Riemann surface whose holomorphic parameterizations w.r.t. j near the negative and positve punctures are $(-\infty, 0] \times S^1$ and $[0, \infty) \times S^1$ (e.g. 2-punctured $\mathbb{CP}^1 \setminus \{0, \infty\} \cong \mathbb{R} \times S^1$),

$$(du - X_{H_\Sigma}(u) \otimes \beta)^{0,1} = 0,$$

in which $\beta \in \Omega^1(\Sigma)$ satisfies $d\beta = 0$ and $\beta = dt$ on cylindrical ends, $du \in C^\infty(\Sigma, T^*\Sigma \otimes u^*TM) = \Omega^1(u^*TM)$, $X_{H_\Sigma}(u) \in C^\infty(\Sigma, u^*TM) = \Gamma(u^*TM)$ is a pullback domain dependent Hamiltonian vector field, which means it is produced from a Riemann surface parameterized Hamiltonian $\{H_z\}_{z\in\Sigma} \in C^\infty(\Sigma, C^\infty(M))$ which is equal to some non-degenerate Hamiltonians H_t's in the cylindrical ends and decays rapidly to zero away from the cylindrical ends in Σ (this can be construct by interpolating with cut-off functions). The ω-compatible almost complex structure is also chosen to be domain dependent, i.e. $\{J_z\}_{z\in\Sigma} \in C^\infty(\Sigma, \mathcal{J}(M,\omega))$, and on the cylindrical ends it is J_t w.r.t. $t \in \mathbb{S}^1$ coordinate only. The "$(0,1)$" stands for the (j,J)-holomorphic part, i.e. $\Gamma(\Lambda_j^{0,1} T^*\Sigma \otimes \Lambda_J^{0,1} u^*TM) = \Omega^{0,1}(u^*TM)$. Now let $\Sigma = \mathbb{CP}^1 \setminus \{0, 1, \infty\}$ the "pair-of-pant", and $H_\Sigma(u)$ is chosen such that near three cylindrical ends it is $H_1, H_2, H_3 \in C^\infty(S^1 \times M)$, then the dimension of the moduli space of the solutions to above Floer equation with asymptotics to x_i's ($1, 2$ are positive asymptotics, 3 is the negative asymptotic) is

$$\dim \mathcal{M}(x_3; x_1, x_2) = \deg x_1 + \deg x_2 - \deg x_3 - n$$

Counting the rigid solutions gives rise to the product operation:

$$\begin{aligned} HF_i(H_1) \otimes HF_j(H_2) &\to HF_{i+j-n}(H_3) \\ x_1 \otimes x_2 &\mapsto \sum_{x_3} \#_{\mathbb{Z}_2} \mathcal{M}^0(x_3; x_1, x_2) x_3 \end{aligned}$$

This actually coincide with the intersection product of the singular homology under the natural isomorphism:

$$\begin{array}{ccc} HF_i(H_1) \otimes HF_j(H_2) & \xrightarrow{\bullet} & HF_{i+j-n}(H_3) \\ \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} \\ H_{i+n}(M) \otimes H_{j+n}(M) & \xrightarrow{\cap} & H_{i+j}(M) \end{array}$$

The energy formula yields:

$$\mathcal{A}_{H_3}(x_3) \leq \mathcal{A}_{H_1}(x_1) + \mathcal{A}_{H_2}(x_2)$$

This implies that product structure is compatible with the filtration, hence:

$$\begin{array}{ccc} HF_*(H_1)\otimes HF_*(H_2) & \xrightarrow{i^a\otimes i^b} & HF_*^{(-\infty,a)}(H_1)\otimes HF_*^{(-\infty,b)}(H_2) \\ \bullet\downarrow & & \downarrow\bullet \\ HF_*^{(-\infty,a+b)}(H_3) & \xrightarrow{\qquad i^{a+b}\qquad} & HF_*(H_3) \end{array}$$

Now, take $H_1 = H$, $H_2 = K$, and $H_3 = H\#K$. Suppose $c(A,H) < a$ and $c(B,K) < b$ s.t. $A \in \mathrm{im}(\Phi_{f,H}\circ i^a)$ and $B \in \mathrm{im}(\Phi_{f,K}\circ i^b)$. Then above commutative diagrams implies that $c(A\cap B, H\#K) \leq c(A,H) + c(B,K)$. This verifies the triangle inequality. In particular, this implies that for $A, B \in H_*(M)\setminus\{0\}$ with $\deg(B) < \dim M$, there is

$$c(A\cap B, H) \leq c(A,H).$$

When $(M,\omega) = (W, d\lambda)$ is a compact Liouville domain, the Hamiltonian Floer homology is also well-defined with an admissible class of Hamiltonian functions $H \in C^\infty(S^1\times\widehat{W})$'s, which satisfy that

$$H|_{(0,\infty)\times\partial W} = h(r)$$

is convex function (i.e. $h''(r)\geq 0$) with $0 \leq h'(r) \leq \tau_0$ and

$$H|_{[R,\infty)} = \tau_0 r + \kappa$$

for some $R > 0$ with $0 < \tau_0 < T_{\min}(\partial W, \alpha)$ and $\kappa \in \mathbb{R}$. This in particular includes non-degenerate $H \in C_c^\infty(S^1\times \mathrm{int}\, W)$. See [FS07] for detail. An important technical ingredient is the choice of *cylindrical* or *contact type* $d\lambda$-compatible almost complex structure: $J \in \mathcal{J}(W, d\lambda)$ satisfies

$$J(r\partial_r) = R_\alpha$$

or equivalently

$$\lambda\circ J = rdr.$$

This provides a C^0-bound of the Floer trajectories in W with the following no-esacpe lemma.

Lemma 2.3. *Let H be an admissible Hamiltonian on a Liouville manifold $(\widehat{W}, d\lambda)$. Let $V \subset \widehat{W}$ be a compact Liouville subdomain, i.e. $\alpha = \lambda|_{\partial V}$ is a positive contact form, and J is clindrical near ∂V and $H = h(r)$ in the cylindrical coordinate (r,x) near ∂V, i.e. $(r,x)\in[1,1+\varepsilon)\times\partial V$. If both asymptotics of a Floer trajectories $u : \mathbb{R}\times S^1 \to \widehat{W}$ are contained in V, then $\mathrm{im}(u)$ is entirely contained in V. When $\{H_s\}_{s\in\mathbb{R}}$ is a smooth family of admissible Hamiltonians such that $\partial_s H_s \leq 0$, this result also holds.*

Proof. We argue by contradiction: assume u escape V. We may assume u transversely intersecting ∂V regarded as $\{r_0\}\times\partial V$ for some $r_0 \in (1, 1+\varepsilon)$, i.e. there is an outward pointing (w.r.t. V in $\widehat{W}$) vector field on ∂V tangent to $\mathrm{im}(u)$ at $\mathrm{im}(u)\cap\partial V$. In the cylindrical coordiante $(r,x)\in[1,1+\varepsilon)\times\partial V$, we have $X_H = h'(r)R_\alpha$ and $\lambda = r\alpha$, and $H = h(r)$ and $\lambda(X_H) = rh'(r)$. The action function in this region reads

$$\mathcal{A}_H(r) = h'(r)r - h(r).$$

Let $S := u^{-1}(\widehat{W} \setminus \operatorname{int} V)$, which is a compact surface with boundary. The Floer equation on $u : S \subset \mathbb{R} \times S^1 \to \widehat{W} \setminus \operatorname{int} V$ is written as

$$(du - X_H(u) \otimes dt)^{0,1} = 0$$

and the energy estimate of $u|_S$ is

$$\begin{aligned} E(u|_S) &= \frac{1}{2} \int_S \|du - X_H(u) \otimes dt\|_J^2 d\mathrm{vol}_S \\ &= \int_S \omega(\partial_s u, J\partial_s u) ds \wedge dt \\ &= \int_S \omega(\partial_s u, \partial_t u - X_H) ds \wedge dt \\ &= \int_S (\omega(\partial_s u, \partial_t u) - dH(\partial_s u)) ds \wedge dt \\ &= \int_S (u^* d\lambda - u^* dH \wedge dt) \\ &= \int_S d(u^*\lambda - H(u)dt) + \int_S H(u) \underbrace{d(dt)}_{=0} \\ &= \int_S d(u^*\lambda - H(u)dt) \end{aligned}$$

By Stokes' theorem:

$$= \int_{\partial S} u^*\lambda - H(u)dt$$

Since $u(\partial S) \subset \partial V$ and $H = h'(r_0)R_\alpha = \lambda(h'(r_0)R_\alpha) = \lambda(X_H)$ on $\{r_0\} \times \partial V$, we have:

$$\begin{aligned} &= \int_{\partial S} u^*\lambda - \lambda(X_H)dt \\ &= \int_{\partial S} \lambda(du - X_H \otimes dt) \end{aligned}$$

Due to $(du - X_H \otimes dt)^{0,1} = \frac{1}{2}((du - X_H \otimes dt) + J \circ (du - X_H \otimes dt) \circ j) = 0$:

$$= \int_{\partial S} -\lambda(J \circ (du - X_H \otimes dt) \circ j)$$

Due to $\lambda \circ J = rdr$ on around ∂V:

$$= \int_{\partial S} -dr \circ (du - X_H \otimes dt) \circ j$$

Due to $dr(X_H) = dr(h'(r)R_\alpha) = 0$ around ∂V:

$$= \int_{\partial S} -dr \circ du \circ j$$

Each $v \in T\partial S$ defining its boundary orientation, so jv points into $\operatorname{int} S$, and $du(jv)$ points outwards of V, and hence we have

$$(dr \circ du \circ j)(v) \geq 0 \qquad \forall v \in T\partial S.$$

This implies that

$$E(u|_S) = -\int_{\partial S} dr \circ du \circ j \leq 0,$$

and since $E(u|_S) \geq 0$, this yields $E(u|_S) = 0$, and therefore

$$du - X_H(u) \otimes dt = 0$$

on S, i.e. $du|_S = X_H \otimes dt = h'(r)R_\alpha \otimes dt$ around ∂V and $R_\alpha \in T\partial V$. This forces $\mathrm{im}(u) \subset \partial V$ and contradicts to the assumption that $\mathrm{im}(u) \not\subset V$.
For H_s, we have in above line "$\int_S d(u^*\lambda - H(u)dt)$" an additional term

$$\int_S \partial_s H_s(u) ds \wedge dt.$$

By putting this term to be ≤ 0, we also eventually get above result. □

The Hamiltonian Floer homology admits also natural isomorphism by continuation maps:

$$H_{*+n}(W, \partial W) \xrightarrow[\cong]{\Phi_{f,H}} HF_*(H) \xrightarrow[\cong]{\Phi_{H,f}} H_{*+n}(W, \partial W)$$

where $f \in C_c^\infty([0,1] \times \mathrm{int}\, W)$ is some C^2-small Morse function. The Hamiltonian spectral invariant $c(A, H)$ are therefore well-defined for any $H \in C_c^\infty(S^1 \times \mathrm{int}\, W)$ and $A \in H_*(W, \partial W)$, and all the statements in Theorem 2.1 hold for $H \in C_c^\infty([0,1] \times \mathrm{int}\, W)$, and in particular $c([W, \partial W], f) = \max f$.

3. Applications in symplectic topology

3.1. Degenerate Arnol'd conjecture.

Theorem 3.1. *For any Hamiltonian diffeomorphism $\varphi \in \mathrm{Ham}(M, \omega)$ where (M, ω) is a closed symplectic aspherical manifold:*

$$\#\,\mathrm{Fix}(\varphi) \geq cl(M) + 1$$

Proof. Assume $\#\,\mathrm{Fix}(\varphi) \leq cl(M)$. Then there exists $A, B \in H_*(M) \setminus \{0\}$ with $\deg(B) < \dim M$ s.t. $c(A \cap B, H) = c(A, H)$ (same reason as for the LS spectral invariant). Let U be any open neighborhood of $\mathrm{Fix}(\varphi)$. Let $f \in C^\infty(M)$ s.t. $f|_{\overline{U}} = 0$ and $f|_{M \setminus \overline{U}} < 0$. Then for any $A \in H_*(M) \setminus \{0\}$ and sufficiently small $\varepsilon > 0$, we construct a smooth family $\{H \# s\varepsilon f\}_{s \in [0,1]}$ producing $\varphi_{H \# s\varepsilon f} = \varphi_H \circ \varphi_{s\varepsilon f}$ which has $\mathrm{Fix}(\varphi_{H \# s\varepsilon f}) = \mathrm{Fix}(\varphi_H)$ for each $s \in [0,1]$, and thus $\mathrm{Spec}(H \# s\varepsilon f) = \mathrm{Spec}(H)$ for each $s \in [0,1]$. Since the action spectrum is measure zero set in $\mathbb{R}$ and $c(H \# s\varepsilon f)$ is continuous in $s \in [0,1]$ by continuity of the Hamiltonian spectral invariant, this implies that $c(H \# s\varepsilon f)$ stays constant as $s \in [0,1]$ varies, so we have

$$c(A, H \# \varepsilon f) = c(A, H).$$

The triangle inequality yields:

$$c(A \cap B, H \# \varepsilon f) \leq c(A, H) + c(B, \varepsilon f) \Rightarrow c(A \cap B, H) \leq c(A, H) + c(B, \varepsilon f)$$

Since $c(A \cap B, H) = c(A, H)$, we deduce that $c(B, \varepsilon f) \geq 0$. On the other hand $c(B, \varepsilon f) \leq 0$ since $f \leq 0$. Thus, $c(B, \varepsilon f) = 0 \Leftrightarrow c(B \cap [M], \varepsilon f) = c([M], \varepsilon f)$. By Theorem 2.1 (4) and Proposition 1.2 (6), this implies that U of $\mathrm{Fix}(\varphi)$ is homologically non-trivial, and leads to $\mathrm{Fix}(\varphi) = \infty$, a contradiction. □

3.2. **Hofer–Zehnder capacity.** Hofer–Zehnder capacity is introduced in [HZ94, §3]. Here, we consider the HZ admissible Hamiltoniians defined in [FGS05, §1], this HZ admissible class of Hamiltonians is equivalent to the original one in [HZ94, §3]. It is easy to show:

$$\pi \leq c_{\mathrm{HZ}}(B^{2n}(1)) \leq c_{\mathrm{HZ}}(Z^{2n}(1))$$

To obtain $c_{\mathrm{HZ}}(Z^{2n}(1)) \leq \pi$, we apply the following important result of bounding Hofer–Zehnder capacity by the displacement energy. The *displacement energy* of an (open) subset $U \subset (M,\omega)$ is defined by

$$e(U;M) := \inf\{\|H\| : \exists \varphi_H \in \mathrm{Ham}_c(M,\omega), \varphi_H(U) \cap U = \varnothing\}$$

If U is non-displaceable, then we put $e(U;M) = \infty$.

Theorem 3.2 ([FGS05, Theorem 1] energy-capacity inequality)**.** *For any subset $U \subset M$ where (M,ω) is a compact symplectically aspherical manifold (when $\partial M \neq \varnothing$, it is a Liouville domain):*

$$c_{\mathrm{HZ}}(U) \leq e(U;M)$$

Proof. We give proof for $(M,\omega) = (W, d\lambda)$ Liouville domain (these arguments carry over verbatim to closed symplectically aspherical manifold), and U is displaceable in $W \subset \widehat{W}$. Suppose $K \in C_c^\infty(S^1 \times \widehat{W})$ generates φ_K which displaces U in $W \subset \widehat{W}$. Suppose $H \in C_c^\infty(\mathrm{int}\, W)$ is an HZ admissible Hamiltonian for $U \subset W$. This means $\mathrm{Per}(H) = \mathrm{Crit}(H)$, and it is actually a C^2-small autonomous Hamiltonian (but probably degenerate in general). By Theorem 2.1 (4), we have

$$c([W,\partial W], H) = \max H.$$

Claim. Given $H, K \in C_c^\infty(S^1 \times \mathrm{int}\, W)$, if φ_K displaces $\mathrm{supp}(H) := \bigcup_{t\in S^1} \mathrm{supp}\, H_t$, then

$$c([W,\partial W], H) \leq \|K\|.$$

Proof of Claim. Pick a smooth function $\chi : [0,1] \to [0,1]$ with $\chi(0) = 0$ and $\chi(1) = 1$, we define $H^\chi(t,x) = \chi'(t)H(\chi(t),x)$. We show that

$$c([W,\partial W], H^\chi) = c([W,\partial W], H)$$

in the following. First, notice that $\varphi_{H^\chi}^t = \varphi_H^{\chi(t)}$, hence $\mathrm{Fix}(\varphi_{H^\chi})$ one-to-one corresponds to $\mathrm{Fix}(\varphi_H)$, and the action values are the same for the corresponding pair of 1-periodic Hamiltonian orbits. Consider

$$\chi_s(t) = (1-s)t + s\chi(t), \qquad s \in [0,1].$$

Then $H^{\chi_0} = H$ and $H^{\chi_1} = H^\chi$. And $\mathrm{Spec}(H^{\chi_s}) = \mathrm{Spec}(H)$ $\forall s \in [0,1]$. By continuity, i.e. Theorem 2.1 (2), $c([W,\partial W], H^{\chi_s})$ is continuous in $s \in [0,1]$, and since $\mathrm{Spec}(H)$ is a measure zero subset in $\mathbb{R}$, the number $c([W,\partial W], H^{\chi_s})$ stay constant as $s \in [0,1]$ varies. Making use of this, we turn H into H^χ s.t. $H^\chi(t,\cdot) = 0$ for $t \in [0, \frac{1}{2}]$, and turn K into $K^{\tilde{\chi}}$ s.t. $K^{\tilde{\chi}}(t,\cdot) = 0$ for $t \in [\frac{1}{2}, 0]$. We further construct a smooth family $\{(sH^\chi)\#K^{\tilde{\chi}}\}_{s\in[0,1]}$. Then $\varphi_{(sH^\chi)\#K^{\tilde{\chi}}} = \varphi_{sH^\chi} \circ \varphi_{K^{\tilde{\chi}}}$. Since $\varphi_{K^{\tilde{\chi}}}$ displaces $\mathrm{supp}(sH^\chi) = \mathrm{supp}(H^\chi)$ for each $s \in [0,1]$, it follows that $x \notin \mathrm{supp}(sH^\chi)$ for every $x \in \mathrm{Fix}(\varphi_{K^{\tilde{\chi}}})$ as well as every $x \in \mathrm{Fix}(\varphi_{(sH^\chi)\#K^{\tilde{\chi}}})$. Therefore, $\mathrm{Fix}(\varphi_{(sH^\chi)\#K^{\tilde{\chi}}}) = \mathrm{Fix}(\varphi_{K^{\tilde{\chi}}})$ and $\mathrm{Spec}((sH^\chi)\#K^{\tilde{\chi}}) = \mathrm{Spec}(K^{\tilde{\chi}})$ for each $s \in [0,1]$. Again by continuity and the discreteness of the action spectrum, we conclude that

$$c([W,\partial W], (sH^\chi)\#K^{\tilde{\chi}}) = c([W,\partial W], K^{\tilde{\chi}}) \qquad \forall s \in [0,1]$$

Notice that $K^{\tilde{\chi}}\#(K^{\tilde{\chi}})^- = 0$,[1] we have

$$c([W,\partial W],H) = c([W,\partial W],H^{\chi}) = c([W,\partial W],H^{\chi}\#K^{\tilde{\chi}}\#(K^{\tilde{\chi}})^-).$$

By triangle inequality (Theorem 2.1 (5)), we have

$$c([W,\partial W],H^{\chi}\#K^{\tilde{\chi}}\#(K^{\tilde{\chi}})^-) \leq c([W,\partial W],H^{\chi}\#K^{\tilde{\chi}}) + c([W,\partial W],(K^{\tilde{\chi}})^-),$$

and

$$c([W,\partial W],K^{\tilde{\chi}}) = c([W,\partial W],K) \leq \int_0^1 \sup_W K_t dt,$$
$$c([W,\partial W],(K^{\tilde{\chi}})^-) = c([W,\partial W],K^-) \leq \int_0^1 \sup_W K_t^- dt,$$

in which

$$\sup_W K_t^- = \sup_W(-K_t\circ\varphi_K^t) = -\inf_W K_t.$$

All together implies

$$c([W,\partial W],H) \leq \int_0^1 \sup_W K_t dt - \int_0^1 \inf_W K_t dt = \|K\|.$$

This finishes the proof of the Claim. □

The Claim implies that for any HZ admissible H of U and any K displacing U:

$$\max H \leq \|K\|$$

The statement follows. □

As a consequence, we can show $c_{\mathrm{HZ}}(Z^{2n}(1)) \leq \pi$ in the following: since $Z^{2n}(1) = B^2(1)\times\mathbb{R}^{2n-2}$ is not bounded subset in $\mathbb{C}^n$, we consider the exhausting sequence of subdomains

$$B^2(1)\times B^{2n-2}(R_1) \subset B^2(1)\times B^{2n-2}(R_2) \subset \cdots \subset B^2(1)\times\mathbb{R}^{2n-2}$$

with $1 \ll R_1 < R_2 < \cdots$ and $R_n \xrightarrow{n\to\infty} \infty$. By Theorem 3.2, we have

$$c_{\mathrm{HZ}}(B^2(1)\times B^{2n-2}(R_n)) \leq e(B^2(1)\times B^{2n-2}(R_n);\mathbb{C}^n) \qquad \forall n\in\mathbb{N}.$$

And one can show that

$$e(B^2(1)\times B^{2n-2}(R_n);\mathbb{C}^n) = e(B^2(1);\mathbb{C}^n) = \pi \qquad \forall n\in\mathbb{N}.$$

This eventually proves:

$$c_{\mathrm{HZ}}(Z^{2n}(1)) \leq e(Z^{2n}(1);\mathbb{C}^n) = \pi$$

[1] $K_t^- := -K_t\circ\varphi_K^t$ generates φ_K^{-1}

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